

Where Does an AI Learn a New Rule?

One-page concept summary · ADAPT · 500–950 words · self-contained briefing (not a project diary).

Paragraph 1 — Design pressure

Machine-learning systems are usually frozen at deployment: they predict well on whatever they were trained on and fail on anything else. Yet humans continually absorb new rules from a handful of examples — a spelling convention, a net-new pattern, a fresh procedure — without "re-training" their brains. The design pressure is to exhibit the same behavior: a model that encounters an unseen task and, given a few demonstrations, solves it *at inference time*, without retraining.

Paragraph 2 — Mechanism

Three families of mechanisms can carry the new rule. **Context adjustment** performs no update at all: the demonstration examples are passed directly with the query, and the model pattern-matches against them. **Parameter adaptation** changes the model's weights at test time (test-time adaptation, TTA): the demonstration set drives gradient updates to θ before readout. **State adaptation** is the middle path a recurrent network uses: persistent weights θ stay fixed ($\Delta\theta = 0$), while an internal adaptive state is updated per example (e.g. $s \leftarrow s + \phi(x,y)$ or a learned recurrent map) and the readout of that state solves the query.

Paragraph 3 — Trade-offs

The three substrates trade off *capacity, cost, retention and interference*. State adaptation gets corrected by causal intervention experiments: destroying the adapted state (swapping it, zeroing it, removing its task component) collapses accuracy, and restoring it recovers accuracy — the state is causally necessary, not decorative. But state capacity is a bottleneck; a lossy compression destroys the statistics the readout needs (a capacity *cliff*, not a slope), and accumulating new rules into a shared register causes $A \rightarrow B \rightarrow A$ forgetting — behaviourally, "catastrophic interference." More test-time compute helps

gradient-based readers monotonically but with diminishing returns.

Paragraph 4 — Research landscape

This is an active frontier. In-context learning (ICL) lets large Transformers pattern-match demonstrations without weight updates (Garg et al. 2022; von Oswald et al. 2023). Test-time training methods retrain at deployment on self-supervision or nearest neighbours and can fix shifted exact figures. Recurrent architectures treat the adaptive state as an emergent "fast weight" memory, trading that finite state against retention. The literature agrees the mechanism you choose determines where the new rule lives, what it costs, and how quickly it is overwritten.

Paragraph 5 — BDH / BDH-CQ frontier connection

Two recent lines are directly relevant. BDH (Kosowski, Uznański, Chorowski & Stamirowska, 2025; arXiv:2509.26507) proposes a post-Transformer, scale-free architecture of locally interacting neuron particles whose *working memory lives in a synaptic state σ^* , updated by a Hebbian rule (in published form, $\sigma \leftarrow \sigma + \eta XY$), separate from persistent parameters. BDH-CQ (Engdahl, Kosowski, Chorowski & Stamirowska, 2026; arXiv:2608.09888) extends this: demonstrations update recurrent memory, and a query is solved by iterative latent computation. The abstract envelope — demonstrations write to state, readout of state solves the query, parameters fixed — is exactly the envelope our transparent toy substrate isolates. We probe the official implementations ourselves and report faithfully whether we can reproduce their headline numbers (see the evidence ledger).

Paragraph 6 — Evidence

In our controlled toy implementations (synthetic linear, symbolic, compositional, and ARC-like grid tasks; exact hidden ground truth): frozen baseline 0.11 $\rightarrow$ state-adapted 1.00 accuracy on unseen linear rules (200 paired tasks, $\Delta\theta = 0$, $\Delta s > 0$); state also=1.00 on symbolic (0.00 frozen). One demonstration suffices only when it identifies the rule; continuous rules need ≥ 2 . Perturb/restore interventions confirm the state carries the task information. The learned-recurrent

checkpoint did **not** acquire the linear rule under our meta-training budget (accuracy $0.09 \approx 0.11$ frozen baseline; learned TTA 0.26 / learned TTT 0.45 stay far below state-adaptation 1.00) — an honest negative, not a reproduction. Limit: toy systems; the classic ARC-like proxy pool is small; BDH-CQ's 150M/29.5% claim was not reproduced by us and stays scoped as PUBLISHED.

Paragraph 7 — Limitation

The capacity probe shows a cliff, not graceful degradation; the recurrent variant under our meta-training budget did not acquire the linear rule; reader expressivity fails on a harder family. Learning-gain claims wait on the full 8–15-participant study (pilot $n=1$ only). No claim is made past toy scale.

Final — Where to continue learning

The interactive laboratory lets you watch the state change per demonstration, break it with interventions, induce interference, and compare with context and parameter adaptation on identical episodes — no retraining, full provenance, every number live or explicitly labelled precomputed/published.

Word count: ~590. Every sentence states a definition, a mechanism, evidence, a limitation, or a connection (per brief guidance).